

“ARTIFICIAL INTELLIGENCE”

Beginnings of Artificial Intelligence

The roots of **Artificial Intelligence** can be traced as far back as ancient times. Greek mythology told stories of mechanical devices and artifacts of varying levels of intelligence, whether authentic or perceived. For example, the myth of Hephaestus included the idea of intelligent robots.

- Scientific research in the field of Artificial Intelligence had no credited founder. It was in the mid to late 1940's, after the Second World War, that recognized work on intelligent machines was undertaken simultaneously by a number of independent scholars. However, many consider **Alan Turing** to be the pioneer. This English mathematician gave a lecture on the topic in 1947 and published his seminal paper *Computing Machinery and Intelligence* in 1950.

- The phrase “Artificial Intelligence” was coined by John McCarthy in 1956. Its initial documented usage was during the Dartmouth Conference, where, for the first time Artificial Intelligence was the sole topic of discussion.

Q. What is Artificial Intelligence?

A. It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable

Intelligence

- The applications of **Artificial Intelligence** are abundant and widespread, especially in developed countries. In fact, Artificial Intelligence has become such a mainstay in today's world that it is taken for granted by the majority of people who benefit from its efficiency. Air conditioners, cameras, video games, medical equipment, traffic lights, refrigerators: all function by way of developments in "smart" technology or **fuzzy logic**. Large financial and insurance institutions rely heavily on Artificial Intelligence to process the huge quantities of **information** that are fundamental to their business practices.

The application of computer speech recognition, though more limited in utilization and practical convenience, has made it possible to interact with computers by using speech instead of writing.

Robotics, the study and development of robots, is another common application whose end goal can be anything from entertainment (such as robot pets), to research (such as Mars rovers), to safety (such as fire detection and extinguishment).

Natural language processing, a subfield of Artificial Intelligence, provides computers with the understanding they require to handle information being encoded by humans. Computer vision instructs computers on how to comprehend images and scenes. It has as some of its goals: image recognition, image tracking and image mapping. This application is valued in the fields of medicine, security, surveillance, military operations, even movie making.

Some more applications of Artificial Intelligence.

- **game playing** You can buy machines that can play master level chess for a few hundred dollars. There is some AI in them, but they play well against people mainly through brute force computation.
- **speech recognition** In the 1990s, computer speech recognition reached a practical level for limited purposes. Thus United Airlines has replaced its keyboard tree for flight information by a system using speech recognition. it is quite convenient. On the **medical diagnosis**, **stock trading**, **robot control**, **law**, other hand, while it is possible to instruct some computers while it is possible to instruct some computers using speech, most users have gone back to the keyboard and the mouse as still more convenient.

understanding natural language Just getting a sequence of words into a computer is not enough. Parsing sentences is not enough either. The computer has to be provided with an understanding of the domain the text is about, and this is presently possible only for very limited domains.

computer vision The world is composed of three-dimensional objects, but the inputs to the human eye and computers' TV cameras are two dimensional. Some useful programs can work solely in two dimensions, but full computer vision requires partial three-dimensional information that is not just a set of two-dimensional views. At present there are only limited ways of representing three-dimensional information directly, and they are not as good as what humans evidently use.

- **expert systems** A ``knowledge engineer'' interviews experts in a certain domain and tries to embody their knowledge in a computer program for carrying out some task. How well this works depends on whether the intellectual mechanisms required for the task are within the present state of AI. When this turned out not to be so, there were many disappointing results. One of the first expert systems was MYCIN in 1974, which diagnosed bacterial infections of the blood and suggested treatments. It did better than medical students or practicing doctors, provided its limitations were observed.
- **heuristic classification** One of the most feasible kinds of expert system given the present knowledge of AI is to put some information in one of a fixed set of categories using several sources of information. An example is advising whether to accept a proposed credit card purchase. Information is available about the owner of the credit card, his record of payment and also about the item he is buying and about the establishment from which he is buying it (e.g., about whether there have been previous credit card frauds at this establishment).

Fuzzy Logic

- Fuzzy logic expands traditional Boolean or classical logic in **order** to allow for partial truths. Classical logic requires that a concept be deemed either true or false, yes or no, black or white. It is a binary based system that does not make allowances for the possibility that the answer may lie somewhere in the middle. Fuzzy logic, on the other hand, is a superset that has been developed to manage the gray areas.

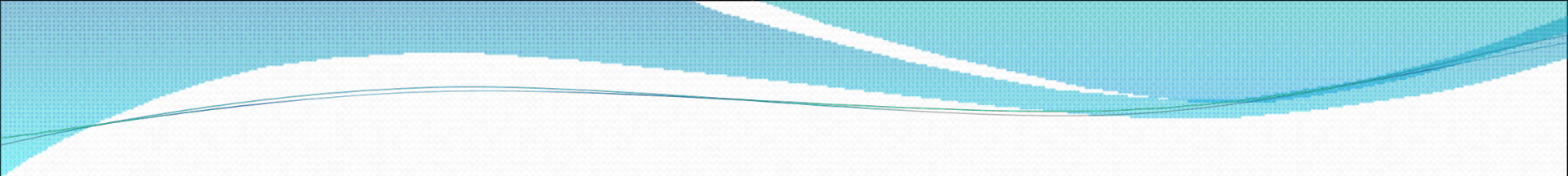
- It is important to note that fuzzy logic operates on definitions that are vague but not random. Furthermore, it would be wrong to equate fuzzy logic to imprecise logic, since it is not the logic, but rather the concept that the logic is handling that is imprecise. By and in itself, the logic is mathematical and organized. For example, the amount of money in a bank account can be quantified, but the amount required to make a man feel wealthy cannot. Therefore, the concept of “wealth” cannot be measured through no fault of the measuring device.

Fuzzy logic normally follows the “if/then” rules of action and reaction. For example, if a temperature reaches the desired setting, then the thermostat switches itself off.

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- Basic applications of fuzzy logic can be found in a growing number of household appliances such as air conditioners, refrigerators, washing machines, security systems, etc.

10 Artificial Intelligence devices.

- Air Conditioners
- Refrigerators,
- Washing Machines,
- Security Systems,
- Cameras,
- Video Games,
- Medical Equipment,
- Traffic Lights,



THE END..